

Advanced (Carolina Reaper)

- Q1.** What is the purpose of a two-way frequency table?
(A) To display one categorical variable
(B) To display two numerical variables
(C) To analyze the relationship between two categorical variables
(D) To calculate the mean of a dataset
(E) To create a bar graph
- Q2.** In a two-way table, what does a relative frequency of 0.25 represent?
(A) A probability of 25%
(B) A count of 25
(C) A percentage of 2.5%
(D) A ratio of 2.5
(E) A proportion of 2.5
- Q3.** If a two-way table has 3 rows and 4 columns, how many cells are in the table?
(A) 6
(B) 10
(C) 12
(D) 7
(E) 9
- Q4.** What type of association is present when the relative frequencies in each row are similar?
(A) Positive association
(B) Negative association
(C) No association
(D) Linear association
(E) Non-linear association
- Q5.** If $P(A) = 0.4$ and $P(B) = 0.3$, what is the probability of both events occurring, $P(A \text{ and } B)$, if they are independent?
- (A) 0.12
(B) 0.40
(C) 0.30
(D) 0.07
(E) 0.50
- Q6.** A researcher creates a two-way table with the variables 'exercise' and 'weight loss'. Which of the following is an appropriate null hypothesis?
(A) Exercise has no effect on weight loss
(B) Exercise always results in weight loss
(C) There is a positive correlation between exercise and weight loss
(D) The relationship between exercise and weight loss is non-linear
(E) There is a negative association between exercise and weight loss
- Q7.** Which of the following can be determined using a two-way frequency table?
(A) The mean of the dataset
(B) The median of the dataset
(C) The mode of the dataset
(D) The relationship between two categorical variables
(E) The standard deviation of the dataset
- Q8.** A two-way table is given with 20 observations in one cell. If the relative frequency of this cell is 0.10, what is the total number of observations in the table?
(A) 100
(B) 200
(C) 150
(D) 50
(E) 80

Open Ended Questions (FRQ)

- Q9.** Use the given two-way table to calculate the relative frequency of each cell and analyze the association between the variables

	Yes	No
Smoker	15	30
Non-smoker	5	50

- Q10.** A researcher collects data on the relationship between the variables 'exercise level' and 'weight loss'. Create a two-way table with 3 rows (low, moderate, high) and 2 columns (yes, no) using the given information: 20 participants with low exercise level had weight loss, 15 participants with moderate exercise level did not have weight loss, 30 participants with high exercise level had weight loss. Calculate the relative frequencies and analyze the association between the variables.

Chef's Tips

- Tip 1: Ensure students understand the concept of relative frequencies and how to calculate them
- Tip 2: Emphasize the importance of analyzing associations between variables in two-way tables
- Tip 3: Remind students to consider the context of the problem when interpreting results